

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

(A State University under Govt. of NCT of Delhi • Established under Delhi Act 06 of 2018)

DEPARTMENT OF INFORMATION TECHNOLOGY



LAB PRACTICAL RECORD FILE PATTERN ANALYSIS & RECOMMENDER SYSTEMS (PARS)

SUBMITTED BY:

Student Name: Monis
Roll Number: 2023UIT3027
Batch & Group: Batch 1 / Group 1
Course / Branch: B.Tech Information Technology

SUBMITTED TO:

Lab Instructor: Dr. Vishal Maheshkar
Department: Department of Information Technology
Institution: Netaji Subhas University of Technology
Campus: Dwarka, Sector-3, New Delhi

Academic Session: 2026–2027

INDEX OF PRACTICAL EXPERIMENTS

Course: Pattern Analysis & Recommender Systems (PARS) Lab | Student: Monis (2023UIT3027)

EXP. #	DATE	EXPERIMENT TITLE & TECHNICAL SCOPE	PAGE	TEACHER'S SIGN.
01	01/08/2026	Study of Datasets Across Different Domains (CIFAR-10 & Iris) Comparative analysis of purpose, source, feature structures (\$d=3072\$ vs \$d=4\$), author performance surveys, and formal citations.	01–06	
02	08/08/2026	Dataset Extraction, Storage, Feature Analysis & Performance Visualization Interactive web ingestion and storage of CIFAR-10 images and Iris dataset with multi-metric evaluation (Accuracy, F1, Log Loss).	07–11	
03	15/08/2026	Multimodal Feature Extraction from Audio Across Timestamps Extracted RMS Energy, Zero Crossing Rate (ZCR), and Spectral Centroid across 10 temporal frames from playable browser audio.	12–15	
04	22/08/2026	Bayesian Classification: Single Feature & Multiple Features Multiple Class Gaussian generative modeling computing Prior, Likelihood, and Posterior distributions with real-time decision boundary plots.	16–19	
05	29/08/2026	NLP Paragraph Preprocessing & Bayesian Domain Classification Text pipeline implementing tokenization, stop word removal, Porter stemming, WordNet lemmatization, and Multinomial Naive Bayes.	20–23	
06	05/09/2026	Bayesian Risk Minimization for Three Classes and Three Actions Conditional risk formulation $R(\alpha_i \mathbf{x}) = \sum \lambda_{ij} P(\omega_j \mathbf{x})$ given an asymmetric operational loss matrix.	24–27	
07	12/09/2026	Linear Regression from Scratch Using Gradient Descent Analytical derivation and manual vectorized update rules ($\Delta w, \Delta b$) predicting housing prices with MSE cost decay curves.	28–31	
08	19/09/2026	Logistic Regression (Binary & Multiclass from Scratch) Sigmoid binary and Softmax multiclass classification from first principles with cross-entropy loss tracking and decision contours.	32–35	
09	26/09/2026	Implementation of Regularization (L1 Lasso & L2 Ridge) Demonstrated shrinkage mechanisms, decision boundary smoothing, and quantitative performance bars before and after regularization.	36–39	
10	03/10/2026	Principal Component Analysis (PCA) with Testing & Reconstruction Derived dominant eigenvector $\mathbf{v}_1$, plotted 1st PC projected line, mapped 1D features, and evaluated test reconstruction errors.	40–43	
11	08/10/2026	Linear Discriminant Analysis (LDA) with Testing & Class Separation Maximization of between-class to within-class scatter $\mathbf{S}_W^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ with projected line, 1D separation, and threshold testing.	44–47	

Complete Practical Source Code & Interactive Applications: <https://github.com/dSaxmonis/pars-lab>